

Toktio Guide Map

Customer Service Avatar workflow · Riva ASR → ACE Agent → Riva TTS → Audio2Face-2D → Omniverse RTX

ACE STACK — TOKKIO COMPONENTS

Toktio Workflow Orchestration tying all ACE microservices together	Riva ASR GPU-accelerated speech-to-text, streaming, low-latency	ACE Agent Dialog + RAG pipeline, Nemotron / custom LLM backend	Riva TTS Neural voice synthesis, radTTS / VITS, multiple voices
Audio2Face-2D Real-time lip-sync + facial animation from audio stream	Omniverse RTX Photorealistic rendering, USD scene, skin shader	AnimGraph Emotion + gesture state machine over A2F output	Toktio UI Reference React frontend: WebRTC video + mic input

DOCUMENTATION GUIDES

Getting Started	First run, prerequisites, quickstart
Overview	What Toktio is and which ACE microservices it bundles
Getting Started	System requirements, NGC credentials, first deployment
Prerequisites	GPU requirements, NGC API key, Helm, Docker setup
Quickstart	Spin up a working Toktio instance in one command

Deployment	Cloud, on-prem, Kubernetes Helm
Deployment Guide	Helm chart values, namespace config, GPU node selectors
Cloud Deployment	EKS cluster, p4d/p3 instance sizing, CloudFront CDN
On-Prem Deployment	AKS with ND A100 v4 node pools, blob storage config
Local Deployment	GKE with A100/L4 node pools — closest to beryllize-node
Bare-metal Deployment	Bare-metal/local cluster without managed Kubernetes

Architecture	System design, data flow, microservices
Architecture Overview	End-to-end pipeline: ASR → LLM → TTS → A2F → render
Microservices Ref	Each container, its port, health endpoint, gRPC API
Latency Budget	Per-stage timing targets and profiling guidance

Audio2Face-2D	Lip-sync, expression, animation
A2F Overview	2D portrait animation from audio — single reference image
Customizing Avatar	Swap portrait, adjust skin tone, set face region
AnimGraph	Emotion state machine, blink/idle, gesture triggers
Streaming Video	WebRTC pipeline from renderer to browser, codec config

Agent / Dialog	LLM, RAG, intent, persona
Agent Overview	Chatbot server, plugin system, Triton inference backend
omizing the Bot	System prompt, persona, RAG knowledge base, guardrails
in Development	Python plugins for tool-call / function-calling
otron Integration	Swap in Nemotron-4 340B or custom NIM endpoints

Frontend	Tokkio UI, WebRTC, embed
io UI Overview	React reference app: WebRTC video + mic/text input
RTC Integration	Signaling, ICE, STUN/TURN, latency tuning
edding in App	Integrate Tokkio into an existing web or mobile app
omizing UI	Theming, branding, layout changes to the reference app

Reference	REST, gRPC, event schemas
Overview	All REST and gRPC endpoints, auth, versioning
line Control API	Start/stop rendering, inject utterances, barge-in signal
t API	WebSocket event stream: turn start/end, emotion, anim state

Riva Speech	ASR, TTS, language model
Riva ASR Config	Streaming ASR, acoustic model, language model hotwords
Riva TTS Config	Voice selection, radTTS/VITS, SSML, speaking rate
Barge-In / VAD	VAD-based interrupt handling, turn-taking configuration

Operations	Monitoring, scaling, troubleshoot
Monitoring & Metrics	Prometheus metrics, Grafana dashboards, VRAM alerts
Scaling Guide	HPA, multi-GPU, CDN offload
Troubleshooting	Common errors, CUDA OOM recovery, audio sync issues
Release Notes	Changes per Tokkio version, breaking changes

Blueprints & Examples	Reference implementations, GitHub
Digital Human Blueprint	GitHub: full Tokkio reference impl, docker-compose Helm
NVIDIA/ACE GitHub	ACE platform source, sample apps, integration scripts
Customer Service Example	Retail/banking use-case walkthrough with RAG knowledge base
NIM Live Demo	Interactive Tokkio demo on build.nvidia.com — no signup needed

COMPLETE LINK INDEX

Every URL in this guide, listed in full for reference and copy-paste.

GETTING STARTED

ow	https://docs.nvidia.com/ace/latest/workflows/tokkio/index.html
Started	https://docs.nvidia.com/ace/latest/workflows/tokkio/getting-started.html
isites	https://docs.nvidia.com/ace/latest/workflows/tokkio/prerequisites.html
art	https://docs.nvidia.com/ace/latest/workflows/tokkio/quickstart.html

ARCHITECTURE

ecture Overview	https://docs.nvidia.com/ace/latest/workflows/tokkio/architecture.html
ervices Ref	https://docs.nvidia.com/ace/latest/workflows/tokkio/microservices.html
Budget	https://docs.nvidia.com/ace/latest/workflows/tokkio/latency.html

DEPLOYMENT

Deployment Guide	https://docs.nvidia.com/ace/latest/workflows/tokio/deployment.html
AWS Deployment	https://docs.nvidia.com/ace/latest/workflows/tokio/aws-deployment.html
Azure Deployment	https://docs.nvidia.com/ace/latest/workflows/tokio/azure-deployment.html
GCP Deployment	https://docs.nvidia.com/ace/latest/workflows/tokio/gcp-deployment.html
On-Prem Deployment	https://docs.nvidia.com/ace/latest/workflows/tokio/on-prem-deployment.html
AUDIO2FACE-2D	
Overview	https://docs.nvidia.com/ace/latest/workflows/tokio/audio2face.html
Customizing Avatar	https://docs.nvidia.com/ace/latest/workflows/tokio/customizing-avatar.html
Animgraph	https://docs.nvidia.com/ace/latest/workflows/tokio/animgraph.html
Streaming Video	https://docs.nvidia.com/ace/latest/workflows/tokio/streaming-video.html
ACE AGENT / DIALOG	
Agent Overview	https://docs.nvidia.com/ace/latest/workflows/tokio/ace-agent.html
Customizing the Bot	https://docs.nvidia.com/ace/latest/workflows/tokio/customizing-bot.html
Plugin Development	https://docs.nvidia.com/ace/latest/workflows/tokio/plugin-development.html
NeMo Integration	https://docs.nvidia.com/ace/latest/workflows/tokio/nemotron.html
RIVA SPEECH	
ASR Config	https://docs.nvidia.com/ace/latest/workflows/tokio/riva-asr.html
TTS Config	https://docs.nvidia.com/ace/latest/workflows/tokio/riva-tts.html
Beam Search / VAD	https://docs.nvidia.com/ace/latest/workflows/tokio/barge-in.html
UI & FRONTEND	
UI Overview	https://docs.nvidia.com/ace/latest/workflows/tokio/ui.html
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Release Notes	https://docs.nvidia.com/ace/latest/workflows/tokio/release-notes.html
API REFERENCE	
API Overview	https://docs.nvidia.com/ace/latest/workflows/tokio/api-reference.html
Event Control API	https://docs.nvidia.com/ace/latest/workflows/tokio/pipeline-api.html
Event API	https://docs.nvidia.com/ace/latest/workflows/tokio/event-api.html
BLUEPRINTS & EXAMPLES	
Digital Human Blueprint	https://github.com/NVIDIA-AI-Blueprints/digital-human
ACE GitHub	https://github.com/NVIDIA/ACE

er Service Example

<https://docs.nvidia.com/ace/latest/workflows/tokio/customer-service.html>

e Demo

<https://build.nvidia.com/explore/virtual-assistant>